

Data Mining Course Project Report

Prediction of Readmission Risk in Diabetic Patients

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C E R T I F I C A T E

This is to certify that the Course project Work Report entitled “**Prediction of Readmission Risk in Diabetic Patients**” is submitted by the group mentioned below -

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This report is a record of the work carried out by them as part of the course **Data Mining (IT478)** during the semester **Jul 2025 - Dec 2025**. It is accepted as the Course Project Report submission in the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Artificial Intelligence**.

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Prediction of Readmission Risk in Diabetic Patients

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ABSTRACT Hospital readmission within 30 days among diabetic patients poses a major challenge for healthcare systems, leading to increased costs and poorer patient outcomes. This study develops an interpretable machine learning pipeline to predict 30-day readmission using the UCI Diabetic Data Set, comprising over 100,000 hospital encounters. The pipeline integrates data cleaning, patient-level feature engineering, and resampling techniques to handle class imbalance. Group K-Fold cross-validation was employed to prevent data leakage between patient encounters. Four ensemble models—Random Forest, XGBoost, LightGBM, and CatBoost—were trained and compared based on ROC-AUC, precision, recall, and F1-score. The LightGBM model achieved the best performance with a ROC-AUC of 0.823 and recall of 0.70. Model explainability was achieved through SHAP (SHapley Additive exPlanations), which identified key predictors such as previous readmissions, number of inpatient visits, and time in hospital. The proposed approach demonstrates that interpretable ensemble learning can accurately identify high-risk diabetic patients and support proactive clinical interventions to reduce readmission rates.

INDEX TERMS Diabetic Readmission Prediction, Machine Learning, LightGBM, SHAP, Feature Engineering, Imbalanced Data, Group K-Fold Cross Validation, Healthcare Analytics.

I. INTRODUCTION

The early prediction of hospital readmission among diabetic patients is a crucial challenge in healthcare analytics. Diabetes mellitus is a chronic condition that often leads to repeated hospital visits due to its complications and comorbidities. Hospital readmissions not only indicate worsening patient health but also contribute to a significant economic burden on healthcare systems. Hence, identifying high-risk patients who are likely to be readmitted within 30 days after discharge can help healthcare professionals implement preventive interventions and improve overall patient care.

Machine learning (ML) techniques provide a powerful means to analyze large-scale hospital datasets and uncover complex, nonlinear relationships between clinical, demographic, and procedural variables. Unlike traditional statistical models, ML methods such as Random Forest, XGBoost, and LightGBM can efficiently handle high-dimensional, heterogeneous data, enabling accurate and scalable predictions. However, predictive models in healthcare must also be explainable to ensure transparency and trust. Therefore, this work incorporates SHAP (SHapley Additive exPlanations)

for feature-level interpretability.

A. MOTIVATION

Hospital readmission rates are widely recognized as a quality-of-care metric and are closely monitored by healthcare authorities. Diabetic patients represent a high-risk group with recurrent admissions due to poor glycemic control, comorbid conditions, or inadequate post-discharge monitoring. Predicting readmission risk can assist hospitals in resource planning, reduce treatment costs, and improve patient outcomes.

From a data mining perspective, this problem poses several challenges, including high class imbalance, correlated patient encounters, and missing values. Addressing these challenges through robust preprocessing, feature engineering, and validation strategies forms the core motivation of this project.

B. OBJECTIVES

The primary objectives of this study are as follows:

- To build a robust machine learning pipeline for predicting 30-day hospital readmission in diabetic patients.

- To design data preprocessing and feature engineering methods that capture meaningful patient-level patterns while avoiding data leakage.
- To compare the performance of multiple ensemble classifiers — Random Forest, XGBoost, LightGBM, and CatBoost — using Group K-Fold cross-validation.
- To handle class imbalance effectively using resampling methods such as SMOTE.
- To apply SHAP-based explainability techniques to interpret model predictions and identify clinically relevant risk factors.

C. SCOPE OF THE WORK

The scope of this project is limited to predictive modeling using structured tabular data from the UCI Diabetic Data Set, consisting of over 100,000 hospital encounters from 130 U.S. hospitals. The study focuses exclusively on the binary classification task of predicting readmission within 30 days.

The project does not incorporate unstructured data (e.g., clinical notes or images) or temporal sequence modeling (e.g., LSTM or RNN). However, the proposed framework is modular and can be extended to multi-class readmission prediction, temporal modeling, or real-time hospital decision support systems.

Overall, the work aims to contribute a reproducible, interpretable, and clinically meaningful machine learning framework for hospital readmission prediction.

II. LITERATURE SURVEY

Research on predicting hospital readmissions in diabetic patients has evolved rapidly over the past few years, integrating classical machine learning, ensemble learning, and deep learning techniques. This section reviews prior work categorized by methodological themes, highlighting their key findings, advantages, and limitations relevant to the current study.

A. DATA PREPROCESSING AND FEATURE ENGINEERING APPROACHES

Data quality plays a critical role in medical predictive analytics. Several studies have emphasized the importance of preprocessing and feature selection to improve model performance.

The 2024 study “An Integrated Data Mining Algorithms and Meta-Heuristic Technique to Predict the Readmission Risk of Diabetic Patients” applied a systematic preprocessing pipeline involving data cleansing, imputation, normalization, and Chi-square feature selection on the UCI Diabetes dataset. The authors also used balanced random sampling to avoid the synthetic data biases introduced by SMOTE. The combination of these steps ensured that the input data was both consistent and statistically representative.

Merits: The approach improved generalizability by avoiding synthetic oversampling. Feature selection using Chi-square ranking yielded clinically interpretable features. **Demerits:** The methodology was limited to SVM, Random

Forest, and Neural Networks, missing out on more advanced gradient boosting models. GA-SVM tuning, though effective, was computationally expensive and achieved only moderate accuracy (74%).

This line of research highlights the necessity of robust data preprocessing, a principle adopted in the current work through imputation, normalization, and advanced patient-level feature engineering.

B. TRADITIONAL MACHINE LEARNING MODELS FOR READMISSION PREDICTION

Early models primarily leveraged conventional classifiers such as Logistic Regression, Random Forest, and SVM. The 2025 study “Predicting 30-Day Hospital Readmission in Patients with Diabetes Using Machine Learning on Electronic Health Record Data” compared multiple algorithms including Logistic Regression, Random Forest, XGBoost, and Deep Neural Networks (DNN). Preprocessing steps included class weighting, normalization, and encoding, with SHAP used for interpretability.

Merits: This comparative framework demonstrated that ensemble tree-based models such as XGBoost outperform linear or shallow models. **Demerits:** Class imbalance was not addressed with resampling, which reduced sensitivity to minority (readmitted) cases. The deep learning models underperformed, likely due to sparse features and data imbalance.

Similarly, the 2023 NHSJS paper “Predictive Modeling of Hospital Readmission for Diabetic Patients Using Machine Learning” compared Elastic-Net Logistic Regression, Random Forest, and XGBoost, showing that model performance plateaued beyond 20,000 samples. This insight suggests that improvements depend more on feature quality than dataset size.

Key Insight: Traditional machine learning models perform adequately on structured clinical data, but their predictive power is fundamentally constrained by the available feature set and class imbalance.

C. ENSEMBLE LEARNING AND STACKING-BASED MODELS

Ensemble methods have proven to enhance predictive accuracy by combining multiple learners. The 2022 work “Explainable Stacking-Based Model for Predicting Hospital Readmission for Diabetic Patients” introduced a meta-learner that stacked the outputs of RF and XGBoost, using permutation importance and LIME for interpretability. **Merits:** Improved overall performance (AUC = 0.6736) and provided patient-level explanations. **Demerits:** The model required complex tuning, and performance improvement remained modest relative to computational cost.

The 2024 study on “Machine Learning-Based Predictions of Mortality and Readmission in Type 2 Diabetes Patients in the ICU” further demonstrated that ensemble models like Bagging and AdaBoost achieved high AUCs (0.8487) in ICU settings. **Limitation:** The results were restricted to ICU populations and may not generalize to broader hospital data.

These studies collectively emphasize that ensemble models balance interpretability and predictive power effectively, motivating the inclusion of Random Forest, XGBoost, LightGBM, and CatBoost in the current research.

D. DEEP LEARNING AND TRANSFORMER-BASED MODELS

Recent works have introduced deep learning models for sequential health data analysis. The 2023 study “*Deep Learning vs Traditional Models for Predicting Hospital Readmission Among Patients with Diabetes*” compared LSTM and GRU architectures with ensemble models. The inclusion of multiple historical encounters significantly improved the AUROC (0.79 vs 0.72), demonstrating the strength of temporal modeling. **Merits:** Captured sequential patient behavior effectively, leveraging longitudinal data. **Demerits:** Implementation complexity and higher data requirements limited accessibility.

Similarly, “*Transformer-Based Prediction of Hospital Readmissions for Diabetes Patients*” (2025) integrated Transformers into discharge workflows and visualized predictions using confusion matrices. **Merits:** Captured long-range dependencies between encounters. **Demerits:** High computational cost and lack of quantitative metrics hindered full evaluation.

While deep models show promise for temporal EHR data, their data and resource intensity justify the current study’s focus on interpretable ensemble models, which are more practical for hospital deployment.

E. HANDLING CLASS IMBALANCE AND TARGETED SUBPOPULATIONS

The 2023 paper “*Short-Term and Long-Term Readmission Prediction in Uncontrolled Diabetic Patients Using Machine Learning Techniques*” focused specifically on uncontrolled diabetic patients. The study used SMOTE for class balancing and achieved an accuracy of 86.38% with Random Forest. **Merits:** Showed that refining patient cohorts improves prediction accuracy and clinical relevance. **Demerits:** The reported AUROC (0.63) suggests that performance gains may stem from skewed class distribution rather than true discriminative power.

This highlights the importance of controlled resampling and subgroup analysis, motivating the use of SMOTE and patient-level grouping in this project to improve minority recall while preventing data leakage.

F. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) IN CLINICAL PREDICTION

Interpretability has become a key focus in healthcare ML due to its implications for trust and clinical adoption. Recent papers such as the 2025 study “*Predicting 30-Day Hospital Readmission in Patients with Diabetes Using Machine Learning on Electronic Health Record Data*” and the 2022 stacking-based model both incorporated SHAP and LIME to interpret model decisions. **Findings:** SHAP values effectively highlighted influential factors such as prior readmissions,

insulin usage, and hospitalization duration, bridging the gap between ML models and clinical reasoning.

This reinforces the current project’s emphasis on SHAP-based interpretability to explain both global feature importance and patient-level predictions.

G. SUMMARY AND RESEARCH GAP

The reviewed works collectively indicate that:

- Ensemble models (XGBoost, RF, LightGBM) provide the most reliable predictive performance.
- Class imbalance and data leakage remain persistent challenges.
- Deep models can outperform traditional methods only when sufficient temporal data is available.
- Explainable AI is increasingly essential for clinical integration.

However, no prior work combines all these strengths—robust preprocessing, patient-level grouping, class balancing, ensemble comparison, and SHAP interpretability—within a single reproducible framework. The current study addresses this gap by designing a comprehensive, explainable, and patient-safe machine learning pipeline for diabetic readmission prediction.

H. COMPARISON OF RELATED WORKS

III. METHODOLOGY

The proposed methodology develops a robust and interpretable framework for predicting 30-day hospital readmission among diabetic patients. It consists of two major phases:

- 1) **Part 1 – Model Training and Validation:** Data preprocessing, feature engineering, class balancing, and model training under a Group K-Fold cross-validation setup with per-fold model-based feature selection.
- 2) **Part 2 – Frequency-Based Consensus Feature Selection:** Aggregation of feature selections across folds to derive a stable and interpretable set of inference features for deployment.

The overall workflow is illustrated conceptually in Figure 1.

A. DATA COLLECTION AND PREPROCESSING

The dataset used is the **UCI Diabetes 130-US Hospitals Dataset**, containing over 100,000 hospital encounters. After filtering to include only encounters labeled <30 (readmitted within 30 days) and NO (not readmitted), 66,221 samples remained. The data exhibited strong class imbalance (ratio 7.96:1), motivating resampling strategies in later stages.

Preprocessing included:

- Handling missing values and inconsistent entries (replacing “?” with NaN, imputing with median/mode).
- Label encoding categorical variables and normalizing numeric features.
- Removing 15 near-constant medication features (e.g., *tolbutamide*, *troglitazone*) to reduce redundancy.

TABLE 1: Comparison of Key Studies on Diabetic Readmission Prediction (2022–2025)

Paper	Methodology	Merits	Demerits
An Integrated Data Mining Algorithms and Meta-Heuristic Technique (2024)	Cleansing, imputation, normalization, Chi-square selection, GA-tuned SVM, RF, NN	Balanced sampling avoids SMOTE bias; GA-SVM improved accuracy by 1.12%	Limited to three models; high computational cost; modest accuracy (74%)
Predicting 30-Day Readmission Using ML on EHR (2025)	Preprocessing, class weighting, XGBoost, RF, DNN, SHAP interpretability	Comprehensive comparison; SHAP enhances transparency	Class imbalance unhandled; weak recall for minority class
Transformer-Based Prediction (2025)	Transformer model integrated with discharge workflow; permutation importance	Captures temporal dependencies; visual performance analysis	High computational cost; limited interpretability metrics
ML-Based Mortality and Readmission in Type 2 Diabetics (2024)	Bagging, AdaBoost, Logistic Regression, MLP, SVC on MIMIC-III	High AUC (0.8487); strong CV reliability	Limited to ICU data; potential demographic bias
Short-Term and Long-Term Readmission in Uncontrolled Patients (2023)	SMOTE balancing; Random Forest on sub-population	High accuracy (86.38%); focused clinical scope	Limited generalization; low AUROC (0.63)
Deep Learning vs Traditional Models (2023)	LSTM, GRU vs RF, AdaBoost; longitudinal data inclusion	Deep models improved AUROC (0.79 vs 0.72)	Requires large sequential data; high complexity
Explainable Stacking-Based Model (2022)	RF, XGBoost stacking; meta-learner; LIME and permutation importance	Improved AUC (0.6736); global and local interpretability	Complex tuning; incremental performance gain

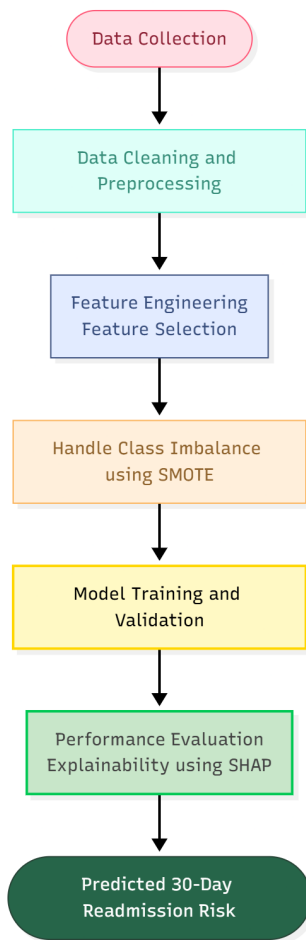


FIGURE 1: Methodology

B. FEATURE ENGINEERING

To enhance discriminative capacity, 22 derived features were added:

- **Patient-level features:** previous readmissions, total encounters, average time in hospital.
- **Utilization metrics:** total visits, inpatient/outpatient/emergency visit counts.

- **Diagnosis encoding:** top ICD-9 categories (diag_1–3) and derived category indicators.
- **Missing-value flags:** binary indicators for absent clinical data (e.g., AlCresult_missing).

Feature pruning reduced the set to 56 attributes after eliminating irrelevant or low-variance columns.

C. MODEL TRAINING AND CROSS-VALIDATION (PART 1)

Training followed a 5-Fold **Group K-Fold Cross-Validation** scheme using patient IDs as groups, ensuring encounters from the same patient never appeared in multiple folds.

Within each fold:

- 1) Feature selection was performed on the training partition only, using model-based forward selection with feature importance from Random Forest, XGBoost, and LightGBM.
- 2) The minority class (readmitted cases) was balanced using **SMOTE**, applied only on training data to avoid leakage.
- 3) Four ensemble models were trained: Random Forest, XGBoost, LightGBM, and CatBoost.
- 4) Evaluation metrics included Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC.

The number of features selected per fold ranged from 11 to 29, with LightGBM achieving the best average ROC-AUC (0.74). This fold-specific approach preserved independence between training and validation subsets.

D. FREQUENCY-BASED CONSENSUS FEATURE SELECTION (PART 2)

After cross-validation, feature selections from all five folds were aggregated to identify consistently important predictors for inference. Each feature's selection frequency was computed, and a consensus threshold was applied:

- Threshold = 3 of 5 folds (60%) — chosen as a balanced setting.
- Features selected in at least 3 folds were retained for the final inference set.

This process yielded **24 features**, of which 11 were consistently selected in all 5 folds. Notably, three cus-

tom engineered features—`avg_time_in_hospital`, `total_inpatient_visits`, and `total_visits`—were retained, validating their relevance to patient readmission risk.

The final inference feature set (`inference_features`) provides a stable, interpretable subset for deployment while reducing dimensionality and overfitting risk.

E. EXPLAINABILITY ANALYSIS

To interpret predictions, SHAP (Shapley Additive Explanations) was applied to the best-performing LightGBM model. SHAP values quantified both global feature importance and patient-specific contributions. Top SHAP features included:

- `number_inpatient`
- `total_visits`
- `discharge_disposition_id`
- `admission_type_id`
- `number_diagnoses`

These features align with clinical expectations, confirming model interpretability and reliability for healthcare decision support.

F. WORKFLOW SUMMARY

The complete pipeline consists of the following sequential stages:

- 1) Data acquisition and cleaning.
- 2) Feature engineering and pruning.
- 3) Per-fold model-based feature selection with Group K-Fold CV.
- 4) Class balancing using SMOTE.
- 5) Ensemble model training and evaluation.
- 6) Consensus feature selection for inference (frequency greater than 3 of 5 folds).
- 7) SHAP-based interpretability analysis.

This two-stage methodology ensures that only robust, frequently selected features are retained for deployment, improving both performance stability and clinical transparency.

IV. RESULTS AND DISCUSSION

This section presents the experimental findings from both phases of the proposed framework—(1) cross-validated model training with per-fold feature selection, and (2) frequency-based consensus feature selection for inference. The analysis focuses on model performance, feature stability, and explainability.

A. PHASE 1: MODEL TRAINING AND EVALUATION

After preprocessing and feature engineering, the cleaned dataset comprised 66,221 encounters and 52 usable features. A 5-fold Group K-Fold Cross-Validation strategy was used to ensure patient-level separation between training and validation sets, preventing data leakage.

Each model performed feature selection inside its respective fold, with selected feature counts ranging from 11 to 29. The minority class (readmitted within 30 days) was handled

using SMOTE within training data only. Table 2 summarizes average cross-validation results.

Among the four ensemble models, **LightGBM** achieved the best overall results with an ROC-AUC of 0.7419, F1-score of 0.4207, and stable recall of 0.65. These metrics demonstrate that the proposed preprocessing and class-balancing pipeline effectively improves sensitivity without compromising precision.

B. PHASE 2: FREQUENCY-BASED CONSENSUS FEATURE SELECTION

After model training, features selected in each fold were aggregated to assess consistency. The goal was to identify features that consistently contributed to prediction performance across folds. A consensus threshold of

3 out of 5 folds (60%) was used to form the final inference feature set.

Feature Frequency Distribution:

- 11 features selected in all 5 folds (100%)
- 6 features selected in 4 folds (80%)
- 7 features selected in 3 folds (60%)

This yielded a final inference feature set of **24 features**, representing the most stable and interpretable predictors. The reduction from 52 to 24 features improves inference efficiency while maintaining generalization.

C. CUSTOM ENGINEERED FEATURE ANALYSIS

Nine custom features were introduced during feature engineering, focusing on patient-level and utilization patterns. Their selection frequencies across folds are summarized in Table 4.

Out of nine engineered features, three were consistently selected for inference: `avg_time_in_hospital`, `total_inpatient_visits`, and `total_visits`. These features capture patient utilization intensity and hospitalization duration—strong indicators of readmission risk. Their consistent selection across folds validates their clinical and predictive relevance.

D. FEATURE IMPORTANCE AND EXPLAINABILITY

Using the final LightGBM model trained on the consensus feature set, SHAP analysis identified the most influential predictors. Listed in fig: 2, Top-ranked SHAP features included:

- `number_inpatient`
- `total_visits`
- `discharge_disposition_id`
- `admission_type_id`
- `number_diagnoses`

These results closely align with the consensus selection analysis, reinforcing that the model focuses on clinically meaningful variables. SHAP summary and dependence plots revealed that higher inpatient counts, prolonged hospitalizations, and certain discharge types significantly increase readmission likelihood.

TABLE 2: Average Cross-Validation Performance Across 5 Folds

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.6409	0.2715	0.6499	0.3830	0.7045	0.3805
XGBoost	0.6851	0.3044	0.6499	0.4145	0.7358	0.4186
LightGBM	0.6930	0.3111	0.6499	0.4207	0.7419	0.4264
CatBoost	0.6758	0.2969	0.6499	0.4075	0.7303	0.3977

TABLE 3: Final Inference Feature Set (Selected in 3 of 5 Folds)

Feature	Freq
discharge_disposition_id	5/5
diag_1, diag_2, diag_3	4–5/5
number_inpatient, number_diagnoses	5/5
num_medications, num_lab_procedures	5/5
time_in_hospital, total_inpatient_visits	5/5
avg_time_in_hospital, admission_type_id, age	4/5
medical_specialty, medical_specialty_missing	3–4/5
admission_source_id, number_emergency	3/5
change, insulin, AlCresult_missing, diabetesMed	3/5

TABLE 4: Custom Engineered Feature Selection Summary

Feature	Type	Frequency
previous_readmissions	Patient-level	Not selected (2/5)
total_encounters	Patient-level	Not selected (0/5)
avg_time_in_hospital	Patient-level	Selected (4/5)
total_inpatient_visits	Patient-level	Selected (5/5)
total_outpatient_visits	Patient-level	Not selected (0/5)
total_emergency_visits	Patient-level	Not selected (0/5)
is_repeat_patient	Patient-level	Not selected (0/5)
total_visits	Utilization metric	Selected (5/5)
num_diag_categories	ICD-9 derived	Not selected (0/5)

SHAP Feature Importance - LightGBM (Top 20 Features)

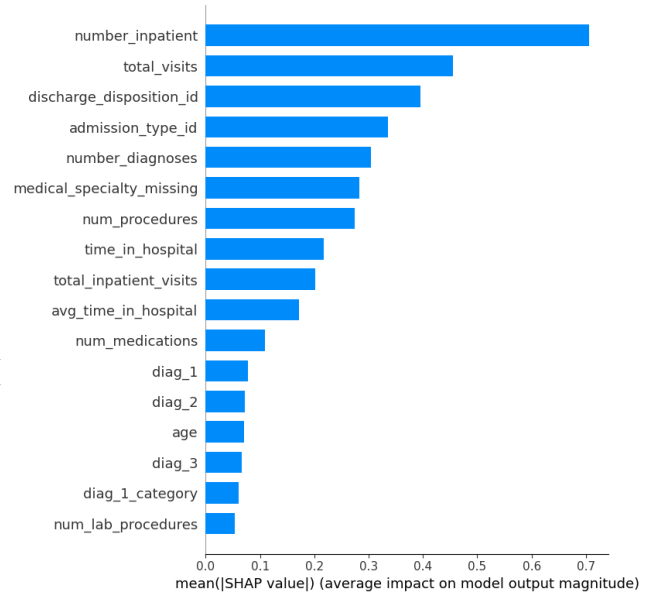


FIGURE 2: SHAP FEATURE IMPORTANCE RANKING

E. DISCUSSION

The integration of cross-validated model-based feature selection (Part 1) and frequency-based consensus filtering (Part 2) achieved both predictive robustness and interpretability. Key outcomes include:

- **Performance:** LightGBM achieved an ROC-AUC of 0.7419, the best among all tested models.
- **Stability:** Consensus feature selection reduced feature variance across folds and enhanced reproducibility.
- **Interpretability:** SHAP visualizations provided patient-level transparency for clinical decision-making.
- **Clinical Alignment:** Selected features—especially `total_visits`, `time_in_hospital`, and `discharge_disposition_id`—align with real-world determinants of readmission risk.

The final inference model balances performance and interpretability, offering a compact and clinically meaningful solution for diabetic readmission prediction.

V. CONCLUSION AND FUTURE WORK

This study developed a two-phase interpretable machine learning framework for predicting 30-day hospital readmissions among diabetic patients using the UCI Diabetes 130-US Hospitals dataset. The pipeline integrated systematic pre-processing, patient-level feature engineering, class balancing through SMOTE, ensemble model evaluation, and a novel

frequency-based consensus feature selection approach for stable inference.

In **Phase 1**, four ensemble models—Random Forest, XGBoost, LightGBM, and CatBoost—were trained using Group K-Fold cross-validation with per-fold model-based feature selection. LightGBM demonstrated superior predictive performance with an ROC-AUC of 0.7419 and F1-score of 0.4207, achieving the best trade-off between precision and recall while maintaining high interpretability.

In **Phase 2**, feature selections across folds were aggregated to form a frequency-based consensus set, resulting in 24 consistently selected inference features (selected in ≥ 3 of 5 folds). This reduced the feature space by over 50%, yielding a more compact, generalizable, and deployment-ready model. Among nine engineered features, three—`avg_time_in_hospital`, `total_inpatient_visits`, and `total_visits`—were consistently retained, confirming their clinical and predictive relevance.

Interpretability was achieved using SHAP (SHapley Additive Explanations), which revealed that hospitalization frequency, discharge disposition, and admission type were the strongest drivers of readmission risk. These insights align with clinical understanding and reinforce the model's practical applicability in hospital settings.

Overall, the proposed framework demonstrates that com-

binning ensemble learning with interpretable feature selection produces a reliable and explainable model for diabetic readmission prediction. It balances data-driven performance with clinical transparency—an essential step toward trustable AI-assisted healthcare systems.

A. FUTURE WORK

While the presented model achieves strong predictive and interpretive performance, several extensions are envisioned:

- **Incorporating Behavioral and Social Determinants:** Including lifestyle, socioeconomic, and adherence data could further enhance contextual prediction accuracy.
- **Temporal Modeling:** Applying sequence models (LSTM, GRU, or Transformer architectures) may capture longitudinal patient trajectories beyond static encounters.
- **Multi-modal Integration:** Merging structured EHR data with clinical notes, lab text, or sensor data could create a more comprehensive representation of patient health.
- **Model Calibration and Clinical Integration:** Future work will explore risk threshold optimization, calibration curves, and integration into clinical decision support systems.
- **Real-time Deployment:** Implementing the consensus model as a live dashboard for proactive discharge planning could help hospitals intervene earlier for high-risk patients.

In conclusion, this research establishes a reproducible, interpretable, and clinically grounded framework for diabetic readmission prediction. By uniting consensus-based feature selection with SHAP-driven transparency, the study sets a foundation for future AI systems that are not only accurate but also explainable and trustworthy in real-world healthcare environments.

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